



OPAL-RT
TECHNOLOGIES



RT-LAB

System Description

Project: CC-221695

Customer: IIT Guwahati

1751 Richardson, suite 1060
Montréal (Québec) Canada H3K 1G6

www.opal-rt.com

Creation - Modification			
Version	Author		Changes Log Book
#	Name	Date	Modification subject
1.0	Mina Hajian	2024-04-18	Initial Release



Contents

Section A - System Summary	6
1 - System Overview.....	6
2 - List of delivered system	6
3 - Login Credentials.....	6
4 - IPMI Credentials	7
5 - Software Requirement.....	7
Section B - System Hardware Description.....	8
1 - OP4512 Real-time Simulator	8
1.1 - General Overview	8
1.2 - Chassis information	8
1.3 - Signal Conditioning Module Configuration	9
Section C - Mapping I/O Blocks to Signal Conditioning in RT-LAB	10
1 - OPAL-RT Controller configuration	10
2 - I/O Bitstream Name and Configuration	11
3 - OP4500 Series I/O Channels assignment.....	12
3.1 - OPT906637 OP4512 IO System	14
3.1.1 - IO Group 1 / Section A (CH 00-31) : Digital Inputs or Outputs	14
3.1.2 - IO Group 1 / Section B (CH 00-31) : Digital Inputs or Outputs	16
3.1.3 - IO Group 2 / Section A (CH 00-15) : Analog Inputs	18
3.1.5 - IO Group 2 / Section B (CH 00-15) : Analog Outputs.....	19
3.1.6 - IO Group 3 / Section A & B (CH 0-5): OP5969-1 RS-422 Digital Input/Output ..	20
Section D - eHS and Schematic Editor	21
1 - Custom eHS Configuration	21



SECTION A - SYSTEM SUMMARY

1 - SYSTEM OVERVIEW

The following image is a diagram showing the layout of the delivered system. It is a top view and is meant to show the component placement and interconnection.

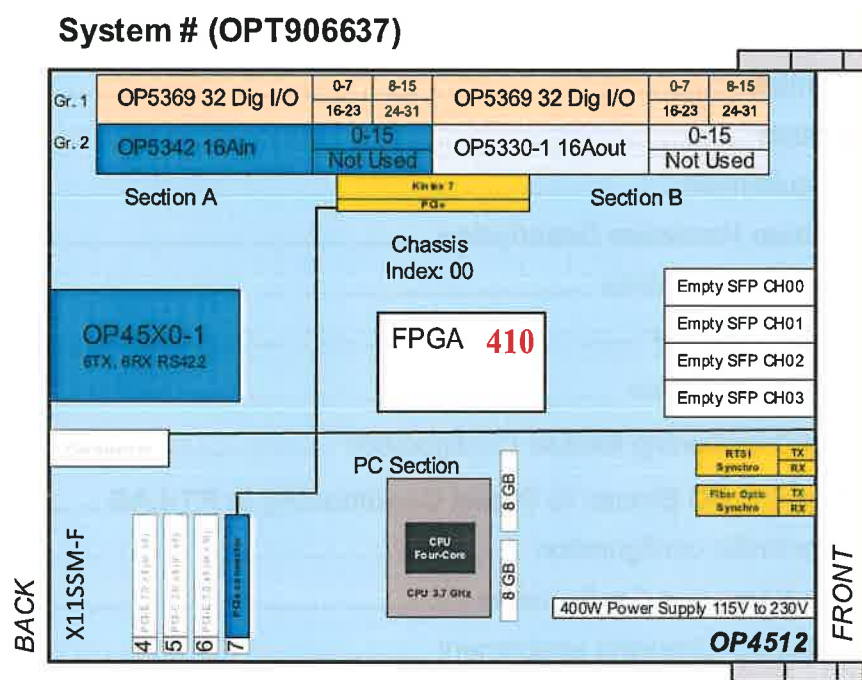


Figure 1 Block diagram of the delivered system

2 - LIST OF DELIVERED SYSTEM

OPT906637 (OP4512 Real-time simulator)

3 - LOGIN CREDENTIALS

The credentials to log on to the real-time simulator are

Username: **root**

Password: **opr935\$**



4 - IPMI CREDENTIALS

The OPAL-RT simulators can be controlled remotely through the IPMI interface. This article of the OPAL-RT knowledge base explains how to do so:

<https://www.opal-rt.com/support-knowledge-base/?article=AA-01708>

Please use the information below to log into the IPMI interface.

Username: ADMIN

Password: MDLDCZFLOQ

5 - SOFTWARE REQUIREMENT

This lists the software that were used for the system at delivery:

MATLAB version : 2021b

RT-LAB version : 2024.1.3.44

eFPGASIM version : 2.17.3.28

Schematic Editor version : 2.17.2.2801

eFPGASIM XSG version : 2.17.3.28

RT-XSG version : 3.5.0.817



SECTION B - SYSTEM HARDWARE DESCRIPTION

1 - OP4512 REAL-TIME SIMULATOR



Figure 2 : OP4512 Real-Time Simulator System

1.1 - GENERAL OVERVIEW

The OP4512 is a complete simulation system operating with a Kintex7 FPGA. It is designed to be used as a desktop (or shelf top) simulator. It contains a powerful Target Computer and a flexible, high-speed Front-End Processor and a signal conditioning stage. The design makes it easy to use with standard connectors (DB37, DB9 or Fiber POF).

The system also offers 6 optional input/output LVDS links that can be used in RS422 or Optical fiber mode. The system delivered is installed with the RS422 option.

In its standard configuration, the lower part of the chassis contains a target computer that can work as a standalone computer.

1.2 - CHASSIS INFORMATION

The target computer delivered includes the features listed in Table 1.

Table 1: List of features

Items	Quantity	Description
Chassis Type	1	OP4512
Operating System		Opal-RT Linux 3.5.4
CPU	1	Intel Xeon E3-1240 V6 Quad-Core 3.7 GHz
Total Core #	4	
Memory	2	8 GB
Motherboard		X11SSM-F-O Supermicro Server Motherboard, Intel socket H4 LGA-1155 µATX
IP Addresses		10.64.117.15 (eno1) – see Figure 3 for identification
AC Input		100-240V, 60-50Hz
FPGA Board Index		00



1.3 - SIGNAL CONDITIONING MODULE CONFIGURATION

The OP4500 series hardware is an entry-level system that contains an FPGA carrier, which can accept four standard OPAL-RT mezzanine boards and a choice of two optional 6TX/6RX transceiver module.

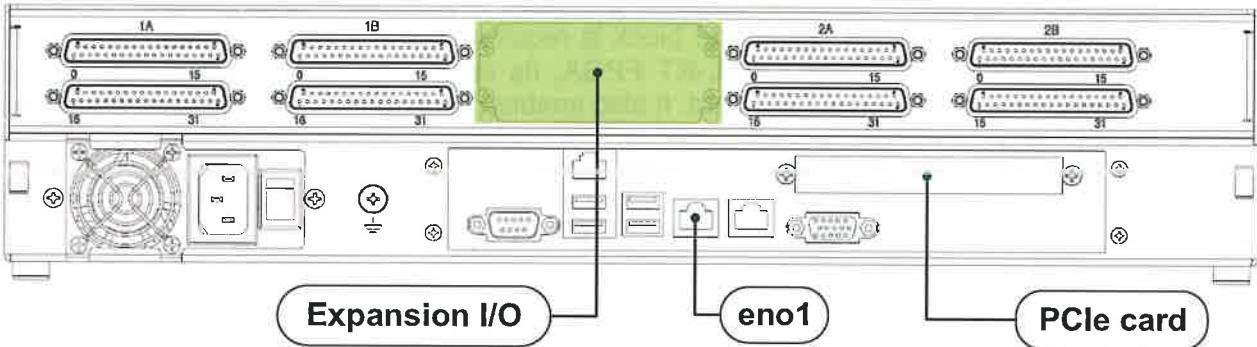


Figure 3: OP4512 V2 Rear View

Similar to other OPAL-RT products, the I/O location is identified by a slot or group number, a module number and a subsection number.

Table 2 lists the conditioning modules provided in the delivered system

Table 2: Signal conditioning module location

I/O Group	Subsection A	Subsection B
1	OP5369 32ch High-Range DIO	OP5369 32ch High-Range DIO
2	OP5342 Analog In (2 MSPS)	OP5330-1 Analog Out
Optional Expansion I/O or Synchro		
OP5969-1 RS422 DIO (6TX, 6RX on Group3)		
Optional PCIe card		
Empty		

RECOMMENDATIONS

Users should adjust the power supply level (through the DB37 connector) to get the proper high voltage level at the DOUT.



Use a proper damping circuit, (a serial resistor capacitor circuit tied to the GND as close as possible to the user Device Under Test) to minimize ringing and over/undershoot according to the connection length (from Digital Output to user device under test).

The following parameters are a good starting point for the RC values: $R=150\Omega$, $C = 100\text{pF}$. Tuning is necessary, according to application parameters.



SECTION C - MAPPING I/O BLOCKS TO SIGNAL CONDITIONING IN RT-LAB

1 - OPAL-RT CONTROLLER CONFIGURATION

To use OPAL-RT I/O modules, an OpCtrl block is required in the RT-LAB model. This block controls the programming of one OPAL-RT FPGA, its initialization and the selection of the hardware synchronization mode of the card. It also enables binding of Send/Recv and I/O blocks to that specific card.

The OpCtrl block can be found in the Simulink block library path:

RT-LAB I/O \ Opal-RT \ Common \ OpCtrl

For the model provided, the block parameters for the 'OpCtrl' block are as follows.

Table 3: IO OPAL-RT Controller parameters

Parameter	Value
Controller Name	'OP4512 OpCtrl'
Board ID	0
Primary Bitstream FileName	TE0741_4-EX-0001-3_5_0_817-eHSx128m_Gen4_DBD8C022QEIO-4F-23.bin
Synchronization mode	Master
Generate External Clock	☐ (Unchecked)
Sample Time (s)	0
Board Type	TE0741
Synchronization type	Optical fiber / Audio
Disable strict hardware mismatch validation	☐ (Unchecked)

IO signals used in the model will be active ONLY if the model is run in Hardware synchronized mode.



2 - I/O BITSTREAM NAME AND CONFIGURATION

A series of I/O tests were run on this demo unit. The I/O test bitstream delivered is described in Table 4.

Table 4: bitstream name and IO configuration

I/O Bitstream	IO Configuration
TE0741_4-EX-0001-3_5_0_817- eHSx128m_Gen4_DBDBD8C022QEiO-4F-23.bin	eHSx64 with • 16 analog outputs • 16 analog inputs • 64 digital inputs or outputs • 2 Quadrature Encoder RS422 inputs /outputs



NOTE:

The eHS license class for this project is eHSx64
Please refer to Section D - eHS and Schematic Editor on page 21 for its configuration.

Bitstream configuration can be checked from RT-LAB, via the Get I/O Infos command.



3 - OP4500 SERIES I/O CHANNELS ASSIGNMENT

The following section is meant to show the relation between the I/O blocks in the RT-LAB model and the signal conditioning modules available on the OP4500 series simulator. They are divided by groups and sections, and each are clearly identified in the illustrations.

The following I/O block mapping is applicable for the hardware test model which is used to validate the hardware functionality and which is provided on the delivery CD.

The following is intended to show the specific pin assignment on standard DB37, DB9 or POF connectors for the system and bitstream delivered.

Table 5: Nomenclature correspondance between Simulink I/O blocks and OP4512 I/Os

Simulink blocks I/O nomenclature	OP4512 I/O ID nomenclature
Slot	Group
Module A	A
Module B	B
Subsection 1 (SS1)	(0-7)
Subsection 2 (SS2)	(8-15)
Subsection 3 (SS3)	(16-23)
Subsection 4 (SS4)	(24-31)

The RT-LAB's Simulink blocks used to access the I/O's in the OP4512's simulator use the following ID nomenclature:

SlotX.ModuleY.SubSectionZ.

In order to map the Simulink ID to the actual physical I/Os in the OP4512 (rear panel) user is advised to use the Nomenclature Correspondence in Table 5. For example, *Slot1. ModuleA. Subsection3* is equivalent to *Group1.A.(16-23)*.



The illustration below shows the arrangement of the subsections of signals on DB37 and DB9 connectors

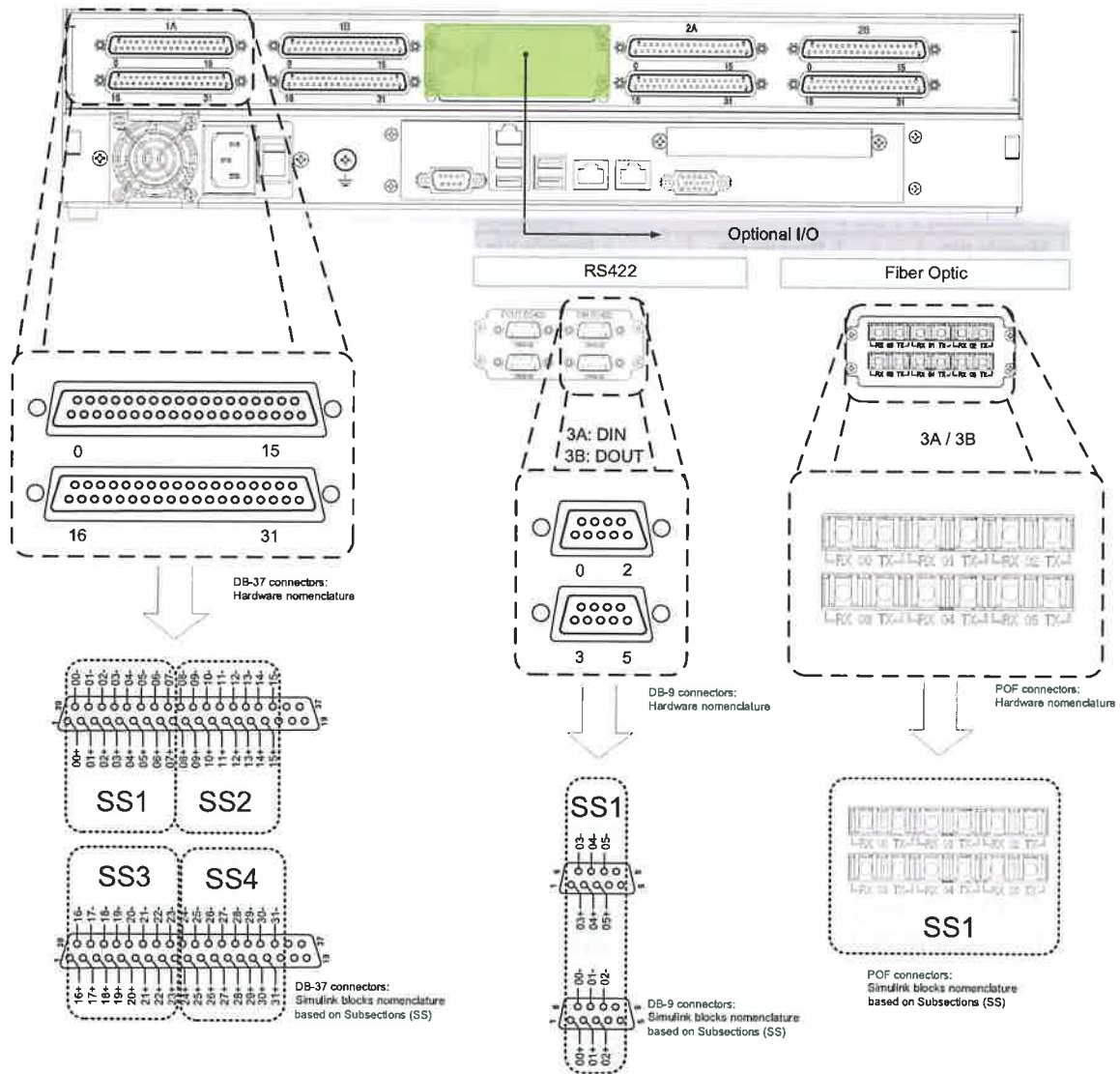


Figure 4: Signals through OP4512 rear connectors

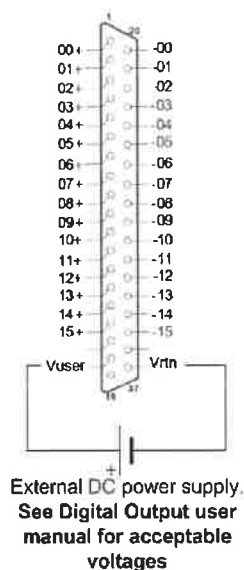


3.1 - OPT906637 OP4512 IO SYSTEM

3.1.1 -IO GROUP 1 / SECTION A (CH 00-31) : DIGITAL INPUTS OR OUTPUTS



Connector P1 Ch. 0-15				Connector P2 Ch. 16-31			
DB37	Module pin assignment	DB37	Module pin Assignment	DB37	Module pin Assignment	DB37	Module pin assignment
1	D00	20	GND	1	D16	20	GND
2	D01	21	GND	2	D17	21	GND
3	D02	22	GND	3	D18	22	GND
4	D03	23	GND	4	D19	23	GND
5	D04	24	GND	5	D20	24	GND
6	D05	25	GND	6	D21	25	GND
7	D06	26	GND	7	D22	26	GND
8	D07	27	GND	8	D23	27	GND
9	D08	28	GND	9	D24	28	GND
10	D09	29	GND	10	D25	29	GND
11	D10	30	GND	11	D26	30	GND
12	D11	31	GND	12	D27	31	GND
13	D12	32	GND	13	D28	32	GND
14	D13	33	GND	14	D29	33	GND
15	D14	34	GND	15	D30	34	GND
16	D15	35	GND	16	D31	35	GND
17	+12V*	36	GND	17	+12V*	36	GND
18	Vuser 1	37	Vrtn 1 (GND)	18	Vuser 2	37	Vrtn 2 (GND)
19				19			



If used as digital output, channels will require a Vuser reference of 4.5 to 24Vdc Max (default internal 4.5 volts), which is provided through the DB37 connector pin 18 and referenced to Vrtn pin 37.

The OP5369 32-channel digital IO board can be configured per 8-channel subsection to be used as digital inputs or digital outputs. The following table is a list of parameters to configure as input or output.

Mezzanine Option block		
Parameters	CH00-07 / Ch08-15 / Ch 16-23 / Ch24-31	
Controller Name	'OP4512 OpCtrl'	
Mezzanine type	OP5369 – 32 DIO	
Direction	Input	Output
Threshold low (volt)	[.470v to 29.4]	N/A
Threshold high (volt)	[.470v to 29.4]	N/A



Digital Output block parameters				
Parameters	CH00-07	CH08-15	CH16-23	CH24-31
Controller Name	'OP4512 OpCtrl'			
DataIn Port Number	11	12	13	14
Number of channels	8	8	8	8
Digital Output Type	- Static DigitalOut - Event Generator - Pulse-Width Modulated Output	- Static DigitalOut - Event Generator - Pulse-Width Modulated Output	- Static DigitalOut - Event Generator - Pulse-Width Modulated Output	- Static DigitalOut - Event Generator - Pulse-Width Modulated Output

Digital Input block parameters				
Parameters	CH00-07	CH08-15	CH16-23	CH24-31
Controller Name	'OP4512 OpCtrl'			
DataOut Port Number	3	4	5	6
Number of channels	8	8	8	8
Digital Input Type	- Static DigitalIn - Event Detector - Pulse-Width Modulated Input	- Static DigitalIn - Event Detector - Pulse-Width Modulated Input	- Static DigitalIn - Event Detector - Pulse-Width Modulated Input	- Static DigitalIn - Event Detector - Pulse-Width Modulated Input

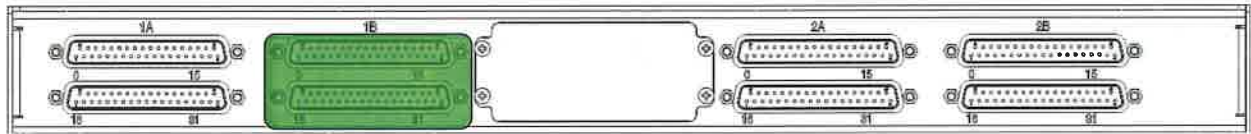
In addition to the parameters listed above, other parameters may need to be configured. More information on these parameters can be found in the documentation of either digital type.

For more information, please visit:

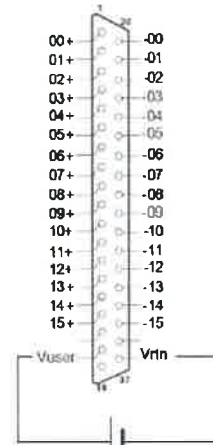
[OP5369 32-Channel Digital IO Board - Hardware Products Documentation - Wiki OPAL-RT](https://wiki.opal-rt.com/display/HDGD/OP5369+32-Channel+Digital+IO+Board)
(<https://wiki.opal-rt.com/display/HDGD/OP5369+32-Channel+Digital+IO+Board>)



3.1.2 -IO GROUP 1 / SECTION B (CH 00-31) : DIGITAL INPUTS OR OUTPUTS



Connector P1 Ch. 0-15				Connector P2 Ch. 16-31			
DB37	Module pin assignment	DB37	Module pin Assignment	DB37	Module pin Assignment	DB37	Module pin assignment
1	D00	20	GND	1	D16	20	GND
2	D01	21	GND	2	D17	21	GND
3	D02	22	GND	3	D18	22	GND
4	D03	23	GND	4	D19	23	GND
5	D04	24	GND	5	D20	24	GND
6	D05	25	GND	6	D21	25	GND
7	D06	26	GND	7	D22	26	GND
8	D07	27	GND	8	D23	27	GND
9	D08	28	GND	9	D24	28	GND
10	D09	29	GND	10	D25	29	GND
11	D10	30	GND	11	D26	30	GND
12	D11	31	GND	12	D27	31	GND
13	D12	32	GND	13	D28	32	GND
14	D13	33	GND	14	D29	33	GND
15	D14	34	GND	15	D30	34	GND
16	D15	35	GND	16	D31	35	GND
17	+12V*	36	GND	17	+12V*	36	GND
18	Vuser 1	37	Vrtn 1 (GND)	18	Vuser 2	37	Vrtn 2 (GND)
19				19			



External DC power supply.
See Digital Output user
manual for acceptable
voltages



If used as digital output, channels will require a Vuser reference of 4.5 to 24Vdc Max (default internal 4.5 volts), which is provided through the DB37 connector pin 18 and referenced to Vrtn pin 37.

The OP5369 32-channel digital IO board can be configured per 8-channel subsection to be used as digital inputs or digital outputs. The following table is a list of parameters to configure as input or output.

Mezzanine Option block		
Parameters	CH00-07 / Ch08-15 / Ch 16-23 / Ch24-31	
Controller Name	'OP4512 OpCtrl'	
Mezzanine type	OP5369 – 32 DIO	
Direction	Input	Output
Threshold low (volt)	[.470v to 29.4]	N/A
Threshold high (volt)	[.470v to 29.4]	N/A



Digital Output block parameters				
Parameters	CH00-07	CH08-15	CH16-23	CH24-31
Controller Name	'OP4512 OpCtrl'			
DataIn Port Number	15	16	17	18
Number of channels	8	8	8	8
Digital Output Type	• Static DigitalOut • Event Generator • Pulse-Width Modulated Output	• Static DigitalOut • Event Generator • Pulse-Width Modulated Output	• Static DigitalOut • Event Generator • Pulse-Width Modulated Output	• Static DigitalOut • Event Generator • Pulse-Width Modulated Output

Digital Input block parameters				
Parameters	CH00-07	CH08-15	CH16-23	CH24-31
Controller Name	'OP4512 OpCtrl'			
DataOut Port Number	15	16	17	18
Number of channels	8	8	8	8
Digital Input Type	• Static DigitalIn • Event Detector • Pulse-Width Modulated Input	• Static DigitalIn • Event Detector • Pulse-Width Modulated Input	• Static DigitalIn • Event Detector • Pulse-Width Modulated Input	• Static DigitalIn • Event Detector • Pulse-Width Modulated Input

In addition to the parameters listed above, other parameters may need to be configured. More information on these parameters can be found in the documentation of either digital type.

For more information, please visit:

[OP5369 32-Channel Digital IO Board - Hardware Products Documentation - Wiki OPAL-RT](https://wiki.opal-rt.com/display/HDGD/OP5369+32-Channel+Digital+IO+Board)
(<https://wiki.opal-rt.com/display/HDGD/OP5369+32-Channel+Digital+IO+Board>)

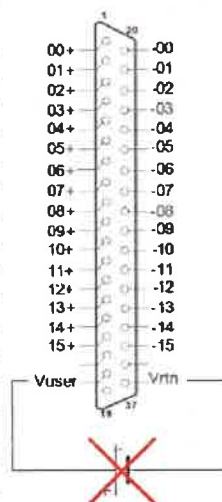


3.1.3 -IO GROUP 2 / SECTION A (CH 00-15) : ANALOG INPUTS



Connector 2A Ch. 0-15

DB37	Module pin assignment	DB37	Module pin Assignment
1	+AIN00	20	-AIN00
2	+AIN01	21	-AIN01
3	+AIN02	22	-AIN02
4	+AIN03	23	-AIN03
5	+AIN04	24	-AIN04
6	+AIN05	25	-AIN05
7	+AIN06	26	-AIN06
8	+AIN07	27	-AIN07
9	+AIN08	28	-AIN08
10	+AIN09	29	-AIN09
11	+AIN10	30	-AIN10
12	+AIN11	31	-AIN11
13	+AIN12	32	-AIN12
14	+AIN13	33	-AIN13
15	+AIN14	34	-AIN14
16	+AIN15	35	-AIN15
17	+12V*	36	DGND
18	Vuser	37	Vrtn 1 A
19			



Do not connect to
external power supply

AnalogIn block parameters

Parameters	CH00-07	CH08-15
Controller Name	'OP4512 OpCtrl'	
DataOut Port Number	7	8
Number of channels	8	8
Sample Time (s)	0	0

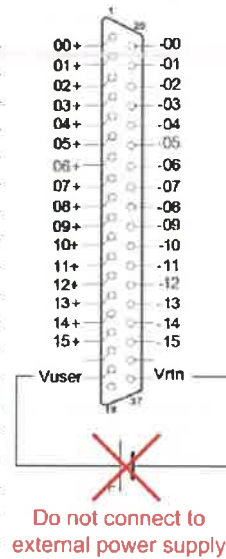


3.1.5 -IO GROUP 2 / SECTION B (CH 00-15) : ANALOG OUTPUTS



Connector 2B Ch. 0-15

DB37	Module pin assignment	DB37	Module pin Assignment
1	+AOUT00	20	CHASSIS_GND
2	+AOUT01	21	CHASSIS_GND
3	+AOUT02	22	CHASSIS_GND
4	+AOUT03	23	CHASSIS_GND
5	+AOUT04	24	CHASSIS_GND
6	+AOUT05	25	CHASSIS_GND
7	+AOUT06	26	CHASSIS_GND
8	+AOUT07	27	CHASSIS_GND
9	+AOUT08	28	CHASSIS_GND
10	+AOUT09	29	CHASSIS_GND
11	+AOUT10	30	CHASSIS_GND
12	+AOUT11	31	CHASSIS_GND
13	+AOUT12	32	CHASSIS_GND
14	+AOUT13	33	CHASSIS_GND
15	+AOUT14	34	CHASSIS_GND
16	+AOUT15	35	CHASSIS_GND
17	+12V*	36	DGND
18	Vuser	37	Vrtn 1 A
19			



NOTE

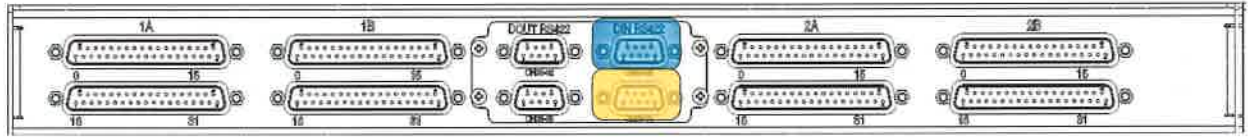
Analog Output channels in this bitstream can be reserved by eHS and managed in the Schematic Editor.

AnalogOut block parameters

Parameters	CH00-07	CH08-15
Controller Name	'OP4512 OpCtrl'	
DataIn Port Number	19	20
Number of channels	8	8
Voltage range	[-16 ... 16]	[-16 ... 16]



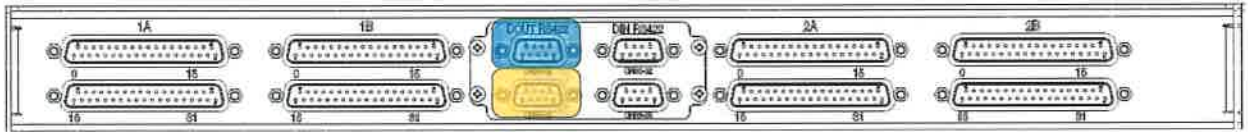
3.1.6 -IO GROUP 3 / SECTION A & B (CH 0-5): OP5969-1 RS-422 DIGITAL INPUT/OUTPUT



Connector 3A DIN RS422 Ch. 00-02			
DB9	Module pin assignment	DB9	Module pin Assignment
1	+ENC0 IN A	6	-ENC0 IN A
2	+ENC0 IN B	7	-ENC0 IN B
3	+ENC0 IN Z	8	-ENC0 IN Z
9			

Connector 3A DIN RS422 Ch. 03-05			
DB9	Module pin assignment	DB9	Module pin Assignment
1	+ENC1 IN A	6	-ENC1 IN A
2	+ENC1 IN B	7	-ENC1 IN B
3	+ENC1 IN Z	8	-ENC1 IN Z
9			

Encoder In Parameters	Values
Controller Name	'OP4512 OpCtrl'
DataOut Port Number	10
Number of Encoder Signals	2



Connector 3B DOUT RS422 Ch. 00-02			
DB9	Module pin assignment	DB9	Module pin Assignment
1	+ENC0 OUT A	6	-ENC0 OUT A
2	+ENC0 OUT B	7	-ENC0 OUT B
3	+ENC0 OUT Z	8	-ENC0 OUT Z
9			

Connector 3B DOUT RS422 Ch. 03-05			
DB9	Module pin assignment	DB9	Module pin Assignment
1	+ENC1 OUT A	6	-ENC1 OUT A
2	+ENC1 OUT B	7	-ENC1 OUT B
3	+ENC1 OUT Z	8	-ENC1 OUT Z
9			

Encoder Out Parameters	Values
Controller Name	'OP4512 OpCtrl'
DataIn Port Number	31
Number of Encoder Signals	2



SECTION D - EHS AND SCHEMATIC EDITOR

The firmware delivered in this project contains an eHS x64 solver that can be used with OPAL-RT Schematic Editor.

[Getting Started - OPAL-RT Schematic Editor Documentation - OPAL-RT \(atlassian.net\)](#)

(<https://opal-rt.atlassian.net/wiki/spaces/PSED/pages/67864689/Getting+Started>)

1 - CUSTOM EHS CONFIGURATION

The system delivered has a custom configuration.

Therefore, the custom firmware must be added to the Schematic Editor database. The required firmware files, including the *.ioconf file, are located in the RTXSG folder of the Customer Delivery DVD.

To set your custom configuration in the Schematic Editor for eHS simulation, please refer to the links below:

[Firmware Manager - OPAL-RT Schematic Editor Documentation - OPAL-RT \(atlassian.net\)](#)

(<https://opal-rt.atlassian.net/wiki/spaces/PSED/pages/67996718/Firmware+Manager>)

[Configuring a Setup for Your Simulation - OPAL-RT Schematic Editor Documentation - OPAL-RT \(atlassian.net\)](#)

(<https://opal-rt.atlassian.net/wiki/spaces/PSED/pages/68093533/Configuring+a+Setup+for+Your+Simulation>)

CONTACT

Opal-RT Corporate Headquarters

1751 Richardson, Suite 1060
Montréal, Québec, Canada
H3K 1G6
Tel.: 514-935-2323
Toll free: 1-877-935-2323

Technical Services

www.opal-rt.com/support

Note:

While every effort has been made to ensure accuracy in this publication, no responsibility can be accepted for errors or omissions. Data may change, as well as legislation, and you are strongly advised to obtain copies of the most recently issued regulations, standards, and guidelines.

This publication is not intended to form the basis of a contract.



OPAL-RT